Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Calculate $3 + 2 \times 4$.
2. Calculate 2^0, 2^3 and 2^{-2}.
3. A population of 80 increases by 5 %. Give the new population.
4. For $f(x) = 3x - 2$, calculate $f(0)$ and $f(4)$.
5. How many integers are there from 3 to 10, endpoints included?
6. Which values does the loop `for k in range(4)` give to k?

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. In 90 seconds, explain the difference between index, term and number of terms. Give a counterexample to a rule guessed from three values.

Investigate before formalising

In pairs, compare two collection offers: 20 kg in the first week then 5 kg more each week; 20 kg then a 20 % increase each week. Conjecture which collects more in total over six weeks, then justify. Allow 10 min for investigation and 5 min for comparison.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.